



Climate Change

041

Understanding and responding to a warming planet

Climate Change studies long-term shifts in global temperature and weather patterns, largely driven by human greenhouse gas emissions, and examines strategies to mitigate and adapt to its effects.

● DEFINITION

Long-term changes in temperature and weather patterns, primarily caused by human activities, especially the burning of fossil fuels.

● WHY IT MATTERS

Climate change affects agriculture, sea levels, weather extremes, and ecosystems worldwide, making it one of the most significant challenges facing society.

● WORKING PRINCIPLE



Climate scientists analyze historical temperature data, greenhouse gas concentrations, and computer models to understand how emissions drive warming and predict future climate scenarios.

● QUICK FACTS

- ▶ Global average temperatures have risen about 1.1°C since pre-industrial times
- ▶ The Paris Agreement, signed by nearly 200 countries, aims to limit warming to well below 2°C

● KEY TERMS

Greenhouse gas Carbon footprint
Net-zero Climate mitigation

● CORE CONCEPTS

- ▶ Greenhouse effect
- ▶ Carbon cycle
- ▶ Climate modeling
- ▶ Mitigation vs. adaptation

● APPLICATIONS

- ▶ Climate policy development
- ▶ Renewable energy planning
- ▶ Disaster preparedness planning
- ▶ Agricultural adaptation strategies

● ADVANTAGES

- ▶ Scientific consensus guides coordinated global action
- ▶ Drives innovation in clean energy
- ▶ Informs disaster preparedness and adaptation

CAREER OPPORTUNITIES

Climate scientist, Sustainability consultant, Renewable energy analyst, Climate policy analyst

RESEARCH AREAS

Climate modeling, Carbon capture technology, Climate adaptation strategies

● REAL-LIFE EXAMPLES

- ▶ International climate agreements like the Paris Agreement
- ▶ Rising sea level impacts on coastal cities
- ▶ Increased frequency of extreme weather events

● LIMITATIONS

- ▶ Global coordination on emissions reduction is politically challenging
- ▶ Climate effects vary greatly by region
- ▶ Long-term nature makes urgency difficult to communicate

INDUSTRY APPLICATIONS

Renewable energy, Government policy, Insurance and risk assessment, Agriculture

FUTURE SCOPE

Climate change is expected to remain a top global priority, driving massive investment in clean energy and adaptation infrastructure.

● CURRENT TRENDS

Accelerating global investment in renewable energy and carbon capture technology to meet net-zero emissions targets.

● SUMMARY

Climate change studies human-driven shifts in global temperature and weather, guiding worldwide efforts to reduce emissions and adapt to impacts.

